

Title: Sex Reassignment at Birth: A Long Term Review and Clinical Implications

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Abstract

This report is a long term follow-up to a "classic" case in the pediatric, psychiatry, and sexology literature. In this case an XY individual had his penis accidentally ablated and was subsequently raised as a female. The initial reports were that this individual was developing into a normally functioning female. The present findings show the individual did not accept this sex of rearing. At puberty this individual switched to living as a male and has successfully lived as such from that time to the present. The significant factors in this switch are presented. In instances of extensive penile damage to infants it is standard to recommend rearing the male as a female. Subsequent cases should, however, be managed in light of this new evidence.

TEXT

Among the more difficult decisions physicians have to make involve cases of ambiguous genitalia or significantly traumatized genitalia. The decision as to how to proceed typically follows the following contemporary advice: "The decision to raise the child as a male centers around the potential for the phallus to function adequately in later sexual relations (pp. 580)."¹ and "Because it is simpler to construct a vagina than a satisfactory penis, only the infant with a phallus of adequate size should be considered for a male gender assignment (pp. 1955)"² These management proposals depend upon a theory which basically says: "It is easier to make a good vagina than a good penis and since the identity of the child will reflect upbringing, and the absence of an adequate penis would be psychosexually devastating, fashion the perineum into a normal looking vulva and vagina and raise the individual as a girl." Such clinical advice, concerned primarily with surgical potentials, is relatively standard in medical texts^{3, 4, 5, 6} and reflects the current thinking of many pediatricians.⁷

This management philosophy is based on two pediatric beliefs held strongly enough that they might be considered postulates: 1) individuals are psychosexually neutral at birth, and 2) healthy psychosexual development is dependent upon the appearance of the genitals. These ideas arise most strongly from the original work of Money and colleagues.^{8, 9, 10, 11, 12} The following are typical pronouncements from that research: "... erotic outlook and orientation is an autonomous psychological phenomenon independent of genes and hormones, and moreover, a permanent and ineradicable one as well (pp. 1397).";⁹ "It is more reasonable to suppose simply that, like hermaphrodites, all the human race follow the same pattern, namely, of psychological undifferentiation at birth."¹⁰ The first postulate was derived, not from normal individuals but from hermaphrodites and pseudohermaphrodites and the second postulate had only anecdotal support. Money no longer holds such extreme views¹³ but his involvement in one particular case was significant enough that it became a totem in the lay press and a classic for the academic and medical community. And, as quoted above, the textbooks have not kept abreast of the new thinking.

Report of a Patient

The case involved a set of normal XY twins, one of whom, John (pseudonym), at seven months of age had his penis accidentally burned to ablation during phimosis repair by cautery.¹¹ After a great deal of debate, the child was seen for consultation at The Johns Hopkins Hospital and, following the two postulates mentioned above, the recommendation was made to unequivocally raise the child as a girl, Joan (pseudonym). Orchiectomy and preliminary surgery to fashion a vagina followed within the year to facilitate feminization. Further surgery was to wait until Joan was older. This management was monitored and reinforced with yearly visits to Hopkins. The treatment was reported as developing successfully and that John was accepting life as Joan.¹¹

"Although this girl is not yet a woman, her record to date offers convincing evidence that the gender identity gate is open at birth for a normal child no less than for one born with unfinished sex organs or one who was prenatally over-or underexposed to androgen, and that it stays open at least for something over a year after birth (pp. 98)."¹²

"The girl's subsequent history proves how well all three of them [parents and child] succeeded in adjusting to that decision (pp. 95)."¹²

The effect of such reports were widespread for theory and practice. Sociology, psychology and women's study texts were rewritten to argue that, as Time magazine (Jan 8, 1973) reported: "This dramatic case . . . provides strong support ... that conventional patterns of masculine and feminine behavior can be altered. It also casts doubt on the theory that major sex differences, psychological as well as anatomical, are immutably set by the genes at conception." Lay and social science writings still echo this case and so do medical texts.^{3, 4, 5, 6, 14} The following quote is typical: "The choice of gender should be based on the infant's anatomy...Often it is wiser to rear a genetic male as a female. It is relatively easy to create a vagina if one is absent, but it is not possible to create a really satisfactory penis if the phallus is absent or rudimentary. Only those males with a phallus of adequate size which will respond to testosterone at adolescence should be considered for male rearing. Otherwise, the baby should be reared as a female."¹⁵

Our current report is in contrast to those reports and advice. It is based on a review of the medical clinical notes and impressions of therapists originally involved with the case and on contemporary interviews. One of us (H.K.S.) was the Head of the Psychiatry Department to which the case was referred in the patients home area. While the patient was assigned to the immediate care of female psychiatrists to foster female identification and role modeling, H.K.S. maintained direct supervisory control of the case. The unique character of this case attracted the attention of the British Broadcasting Co. and they invited M.D. as a consultant.¹⁶ In 1994 and 1995, we collaboratively reinterviewed and recorded John, his mother and his wife to provide updated accounts of his progress. Findings are listed in general chronological order under the more appropriate of the two postulates for pediatric sexual assignment. John himself, while desiring to remain anonymous, strongly desires his case history be made available to the medical community to reduce the likelihood of others having his psychic trauma.

Postulate 1: Individuals are psychosexually neutral at birth.

Mother recalls: "As soon as he had the surgery, the doctor said I should now start treating him as a girl, doing girl things and putting him in girl's clothes. But that was a disaster. I put this beautiful little dress on him, ... and he [immediately tried] to rip it off... I think he knew it was a dress and that it was for girls and he wasn't a girl."

On the other hand, Joan could act quite feminine when she wanted to, and was reported as doing so, e.g. mother was quoted to have said: "One thing that really amazes me is that she is so feminine. I've never seen a little girl so neat and tidy as she can be when she wants to be... (pp. 119)."¹¹ However, she most often, would prefer to reject such behavior. It was also more common that she, much more than the twin brother, would mimic Father. One incident Mother related was typical: When the twins were about 4 or 5 they were watching their parents. Father was shaving and Mother applying makeup. Joan applied shaving cream and pretended to shave. When Joan was corrected and told to put on lipstick and makeup like Mother, Joan said: "No, I don't want no makeup, I want to shave."

Girl's toys, clothes and activities were repeatedly proffered to Joan and most often rejected. Throughout childhood Joan preferred boy's activities and games to those of girl's; she had little interest in dolls, sewing or girl's activities. Ignoring the toys she was given, she would play with her brother's toys. She preferred to tinker with gadgets and tools and dress up in men's clothing; take things apart to see what makes them tick. She was regarded as a tomboy with an interest in playing soldier. Joan did not shun rough and tumble sports nor avoid fights.

John recalls of Joan at the age of 12 or 13 wanting an umbrella:

"I had a couple of bucks and went to the store to take a look at the umbrellas, and right beside the umbrellas was the toy section. I started to eyeball a machine gun. I said to myself 'Do I have enough money for that?'... I put the gun on the counter and asked the clerk if I had enough money. She had that look like 'You don't have enough but we'll let you go anyway.' I used it to play army with my brother". Brother often refused to let Joan play with his toys, so she also saved her allowance money and bought a truck of her own.

Joan's realization that she was not a girl jelled between ages 9 and 11 years. John relates:

"There were little things from early on. I began to see how different I felt and was, from what I was supposed to be. But I didn't know what it meant. I thought I was a freak or something; ... I looked at myself and said I don't like this type of clothing, I don't like the types of toys I was always being given, I like hanging around with the guys and climbing trees and stuff like that and girls don't like any of that stuff. I looked in the mirror and sees my shoulders are so wide, I mean there is nothing feminine about me. I'm skinny, but other than that, nothing. But that is how I figured it out. [I figured I was a guy] but I didn't want to admit it, I figured I didn't want to wind up opening a can of worms."

Joan knew she already had thoughts of suicide brought on by this sort of cognitive dissonance and didn't want additional stress. Joan fought both the boys as well as the girls who were always "razzing" her about her boy looks and her girl clothes. She had no friends; no one would play with her. "Every day I was picked on, every day I was teased, every day I was threatened. I said enough is enough ..." Mother relates that Joan was good looking as a girl. But it was "When he started moving or talking, that gave him away and the awkwardness and incongruities became apparent."

The other girls teased Joan so aggressively that she felt forceful retaliation was called for. One girl sat behind Joan and continued to hit her. "[John demonstrating] I grabbed her like that, by the shirt, and rammed her round the wall like this, threw her on the ground... until the teacher grabbed me." This resulted in Joan being expelled from school.

Despite the absence of a penis, Joan often tried to stand to urinate. This made a mess as it was difficult to direct the urine stream. While she learned to sit and void she would nevertheless continue to occasionally stand and urinate. Despite admonitions against the behavior and the untidiness, Joan persisted to such an extent that, at school, she was caught standing to urinate in the girls' bathroom

sufficiently often that the other girls refused to allow her entrance. Mother recalls the other girls threatening to "kill" her if she persisted. Joan would also go to the boy's lavatory to urinate.

Joan was put on an estrogen regimen from the age of 12 years but rebelled against taking the hormones. They made her "feel funny" and she didn't want to feminize. She would often dispose of her daily dose. She unhappily developed breasts but wouldn't wear a bra. Things came to a head at the age of 14. In discussing her breast development with her endocrinologist she confessed "I suspected I was a boy since the second grade". The physician, who personally believed Joan should continue to take her estrogens and proceed as a girl, used that opening to explore in a nonjudgmental manner, the possible male or female paths available and what either one would mean. Since the local management team had already noticed Joan's preference for boys' activities and refusal to accept female status and they had discussed among themselves the possibility of accepting Joan's change back to male, the endocrinologist explored Joan's options with her. Shortly thereafter, at age 14 years, Joan decided to switch to living as a male.

Joan was the daily butt of her peer's jibes and the local therapists, having knowledge of her previous suicidal thoughts, went along with the idea of sex re-assignment. In a tearful episode following John's prodding, his father told him of the history of what had transpired and why. John recalls: "All of a sudden everything clicked. For the first time things made sense and I understood who and what I was."

John requested "male hormone" shots and gladly took these. He requested a mastectomy and phalloplasty. The mastectomy was completed at the age of 14 years; surgeries for phallus construction were at age 15 and 16 years. After the surgical procedures, John adjusted well. As a boy he was relatively well accepted and popular with both boys and girls. At 16 years, to attract girls, John obtained a windowless van with a bed and bar. Girls, who as a group, had been teasing Joan, now began to have a crush on John. When occasions for sexual encounters arose, however, he was reluctant to move erotically. When he told one girl friend why he was hesitant --that he was insecure with his penis-- she gossiped at school and this hurt John very much. Nevertheless, his peers quickly rallied round and he was accepted and the girl rejected.

John's life subsequently was not unlike other boys with an occult physical handicap. Subsequent to his return to male living he felt his attitudes, behaviors and body were in concert; they weren't when living as a girl. At the age of 25 he married a woman several years his senior and adopted her children.

Postulate 2: Healthy psychosexual development is intimately related to the appearance of the genitals.

First in Baltimore, and then with the local therapists prior to the sex reassignment, Joan's expressed feelings of not being a girl would draw ridicule and she would be told something like: "All girls think such things when they are growing up". He recalls thinking: "You can't argue with a bunch of doctors in white coats; you're just a little kid and their minds are already made up. They didn't want to listen." To ease pressures to act as a girl, Joan would often not argue or fight the assignment and would "go along".

Starting at age 7 years, Joan began to rebel at going for the consultations at the Johns Hopkins Hospital. Her reason was discomfort and embarrassment with forced exposure of her genitals and constant attempts, particularly after the age of 8 years, to convince her to behave more as a girl and accept further vaginal repair. This was always strongly resisted and led to recurrent confrontations. To encourage the visits and temper Joan's reluctance to travel to the consultants, they were combined with vacation trips.

In Baltimore, the consultants enlisted male-to-female transsexuals to convince Joan of the advantages of being female and having a vagina constructed. She was so disturbed by this that in one instance Joan, at age 13 years, ran away from the hospital. She was found hiding on the roof of a nearby building. After age 14 years, Joan adamantly refused to return to the hospital. Joan then came instead fully under the care of local clinicians. This group consisted of several pediatricians, 2 pediatric surgeons, an endocrinologist, and a team of psychiatrists.

John recalls thinking, from preschool through elementary school, physicians were more concerned with the appearance of Joan's genitals than was Joan. Her genitals were inspected at each visit to The Johns Hopkins Hospital. She thought they were making a big issue out of nothing and they gave no reason to think otherwise. John recalls thinking: "Leave me be and then I'll be fine ... Its bizarre. My genitals are not bothering me, I don't know why its bothering you guys so much."

When asked what Joan thought of her genitals as a youngster, John replied "I didn't really have anything to compare myself against other than my brother when we were taking a bath." Mother confirmed that as a devout family in a very conservative religious community there would be few opportunities for the twins to have seen anyone else's genitals. Nudity was never acceptable. At their yearly visit to The Johns Hopkins Hospital the twins were made to stand naked for inspection by groups of clinicians and to inspect each other's genitalia. This experience, in itself, was recalled with strong negative emotions. John's brother, decades later, recalls the experience with tears.

John recalls frustration, which remains, at not having his feelings and desires recognized. Without consideration of genitals, with the obvious absence of a penis, he nevertheless knew he was not a girl. When he tried to express such thoughts: the doctors "would change the subject whenever I tried to tell [doctor] how I felt. [He] didn't want to hear what I had to say but wanted to tell me how I should feel." Clinical notes from the time report he felt "like a trapped animal".

In middle school Joan had a very difficult time making friends. Her clothes and demeanor, to her peers, did not jibe. Because of her behaviors they teased her with names like: "caveman" and "gorilla". Few children would play with her. None of Joan's peers knew anything of her genitals.

At first the local physicians, as did Joan's parents, continued to reinforce her feminine identity as suggested by the consultants and continued to treat Joan as a girl preparing her for vaginal reconstructive surgery and life as a female. Psychotherapy, primarily by female therapists, aimed to reinforce her female identity and redirect her male ideation. This became increasingly difficult due to Joan's growing insistence she did not see herself as a girl and anger at being treated as one. John recalls: "They kept making me feel as if I was a freak."

John knew what the clinicians wanted and recognized it wasn't what he wanted. Starting at age 14, against the recommendations of the clinicians and family, and without yet knowing of the original XY status, Joan, as much as possible, refused to live as a girl. Jeans and shirt, due to their gender neutral status, were her usual preferred clothes; boys games and pursuits her usual activities. Joan's daytime fantasies and night dreams during elementary school involved seeing herself "as this big guy, lots of muscles and a slick car and have all kinds of friends..." She aspired to be a mechanic. She rejected requests to look at pictures of nude females he was supposed to emulate. Rorschach and Thematic Apperception Tests at the time elicited responses more typical of a boy than a girl. Her adamant rejection of female living and her improved demeanor and disposition when acting as a boy, convinced the local therapists of the correctness of sexual re-assignment.

Following the surgery for penile construction there was difficulty with urethral closure and despite repeated attempts at repair, that problem was never rectified. John now urinates through a fistula at

the base of his penis while sitting down. Much of the penis is without sensation as are the areas of scarring from where the grafts were taken.

John's first sexual partner was a girl. He was 18 years old. While living as a girl and afterward as a boy, John was approached sexually by males. He claims never to have been attracted to any and his responses to such questions were matter-of-fact and not homophobic. John thinks his first recognizable sexual interest occurred about the age of 17 or 18 years although he does recall wanting to go see the "sexy" Rockettes in New York on one of his trips to see the consultants.

Coitus is "occasional" with his wife. This frequency is sufficient for his needs but is less than his wife would desire. They mostly pleasure each other with a great deal of physical affection and mutual masturbation. John can have coital orgasm with ejaculation.

John recalls thinking it was small minded of others to think all his personality was summed up in the presence or absence of a penis. He expressed it thus: "Doctor ... said, it's gonna be tough, you're going to be picked on, you're gonna be very alone, you're not gonna find anybody unless you have vaginal surgery and live as a female. And I thought to myself, you know I wasn't very old at the time but it dawned on me that these people gotta be pretty shallow if that's the only thing they think I've got going for me; that the only reason why people get married and have children and have a productive life is because of what they have between their legs." ... "If that's all they think of me, that they justify my worth by what I have between my legs, then I gotta be a complete loser."

General comments

As an adult, John was asked "Why not accept being a female rather than fighting it?" His answer was simple. Basically he wanted to please his parents and placate the doctors so he often went along. But doing so didn't feel right and the confusion between his feelings and theirs he saw was mentally devastating and would lead to suicide if he were forced to continue. The most often voiced and deeply felt emotion expressed by Joan was always feeling different from what was expected or desired by others. At first, as a toddler, the feeling of being different was relatively amorphous. Then, even as a preschooler, it shifted to clearly being different from girls. And later, in elementary school, he began to not only feel different from girls but similar to the group called "boys". Certainly having a twin might have made this comparison much easier. Such a progression in thinking is common for atypical individuals such as homosexual males and females¹⁷ and hermaphroditic individuals or those with ambiguous genitalia.¹⁸

The transition was gradual. When Joan thought she might really be a boy, instead of the girl her parents and the doctors told her she was, the psychic discord greatly frightened her even though she had suspected such was true since second grade. When finally told the truth, she was relieved since her feelings now made sense. John's anger at not being told the truth from the beginning persists.

Following John's sex re-reassignment, the family decided to disregard the clinical recommendation to move from their family home. Instead they stayed and were open about the change. Aside from the financial concerns, the parents decided the word would get out anyway. This strategy seemed to work and John was accepted in a way that Joan never was.

John was given testosterone following his return to male status. As is typical of many teen-age boys, John began to work out with weights. He blossomed into an attractive muscular young man.

According to John's wife: "Before he came along I was a lot tougher on the kids because I had to be. [Now] John is the real hard one and I am the soft one. ... There is no doubt who wears the pants in this

family."

John is a mature and forward looking man with a keen sense of humor and balance. While still bitter over his experience, he philosophically accepts what happened and is trying to make the most of it with support from his wife, parents and family. He has job satisfaction and is generally self-assured.

Comment

Long term follow-ups of case reports are unusual but often crucial. This up-date to a case originally accepted as a "classic" in fields ranging from medicine to the humanities completely reverses the conclusions and theory behind the original reports. Cases of infant sex reassignment require inspection and review after puberty; 5 and even 10 year post sex reassignment follow-ups are still insufficient.

Possibly the initial impressions of the consultants^{11, 12} were appropriate at the time and Joan's behavior shifted with development. However, it appears from clinical notes and impressions of the local physician's at the time, and John's contemporary recollections, that at no time was he fully accepting the sex reassignment as a girl. Indeed, the local physicians expressed their reservations early on.¹⁶ When it became obvious, however, that the original management program of maintaining this male as a girl was no longer tenable and proving psychologically damaging --even life threatening-- they revised their thinking.

It is also possible that interpretations from the early years were mistaken since it is often difficult to see results not in keeping with one's hypotheses and management plans. Behavioral manifestations of typical boy behaviors would thus repeatedly be interpreted as tomboyish. This was true regardless if this was in terms of preferred activities, games, toys or clothing. Significantly, the conclusions that hermaphrodites and pseudohermaphrodites offer a model for normal development had been challenged before.¹⁹⁻²⁴ The implications of such challenges does not seem to have been accepted or integrated by the majority of pediatricians or surgeons.⁷

John was repeatedly admonished for behaving like a boy. Such management is in keeping with the belief²⁵⁻²⁸, that any acquiescence to doubt expressed by the patient would decrease the likelihood of an eventual successful outcome. It is circular reasoning to contend that Joan did not accept the imposed sex of rearing because of ambiguity in treatment. There is no evidence of such and the initial reports held that the rearing was appropriate.^{11, 12} It is known, particularly from transsexuals, that casting doubt as to sexual identity usually forces greater introspection and security regarding a preferred life's direction even though counter to upbringing, parent's wishes, social and cultural norms and may result in less than adequate genitalia.^{29, 30}

In the case under consideration, the initial management protocol was predicated on postulates which saw successful treatment involving attention to a male's self-image supposedly dependent upon the adequacy of a functional penis. While it can be accepted that such adequacy is important there is no body of data establishing its centrality.

Other considerations are in order. Gender reassignment as proposed for John, and the postulates upon which they are based, assume the individual will learn to accept rearing-appropriate sex-typical behaviors, particularly when the genitals are at issue. These situations range from the urinary to erotic to narcissistic. Such behaviors, while important, are only one aspect of an individual's total sexuality. An individual's sexual profile is comprised of at least five levels: gender Patterns, Reproduction, sexual Identity, Mechanisms and sexual Orientation (PRIMO).^{3, 32}

The sex reassignment of John to Joan only attended to the gender patterns, and gender roles to which he would be subject with the expectations his identity and other levels would follow. Joan did indeed become aware of the social expectations concomitant with the female gender but these were not in keeping with those with which he felt comfortable. Standing to urinate, despite its housekeeping and social consequences is a dramatic display of preference. The sex reassignment thus obviously failed in the area in which it was most designed to succeed.

But it failed in the other four levels as well. The contrast between the female gender-typical behaviors the child was being asked to accept and his inner directed behavior preferences presented a discordance that demanded resolution. Joan's analysis of the situation was that she best fit in, not as a girl but as a boy. Thus, despite her upbringing, Joan's sexual identity developed as a male. Sex reassignment also obviously went against Joan's or John's reproductive character. Castration removed any reproductive capacity. Certainly unaware of this as a child, John very much resents this now and decries this loss. Castration also removed the androgen source for sex-typical mechanisms of sexual arousal and other physiological processes. His ability to ejaculate returned with androgen treatment. The castration and surgical scarring, however, has dramatically reduced erotic sensitivity to the perineum and subsequently reduced this option. And significantly, as many studies strongly indicate, sexual orientation is prenatally organized or at least predisposed.³³⁻⁴⁰ The sex reassignment did nothing to effect sexual orientation. Joan remained totally gynecophilic despite being reared as a girl.

Comments from John's parents reveals another important consideration. With a sex reassignment they are asked to make a dramatic psychological adjustment in how to rear an otherwise normal child. Mother herself required psychiatric treatment to help manage her feelings in this regard. The penile ablation, notwithstanding, they were more comfortable dealing with their child's original sex and the accident than with a reassigned sex. Although they had definitely tried to make a success of the original sex reassignment, they were very supportive --while guilt ridden-- of Joan's reversion to John.

This last decade has offered much support for a biological substrate for sexual behavior. In addition to the genetic research mentioned above there are many neurological and other reports which point in this direction.^{3, 32, 41-56} The evidence seems overwhelming that normal humans are not psychosexually neutral at birth but are, in keeping with their mammalian heritage, predisposed and biased to interact with the environment, familial and social forces, in a male or female mode. This "classic" case demonstrates this dramatically. And the fact that this was particularly expressed at puberty --a critical period-- is logical and has been predicted.^{20, 44}

Although this report is of the classic case of sex reassignment so often cited in the literature, follow-ups to other related cases are available. Reilly and Woodhouse⁵⁷, described²⁰ patients with micropenis reared as boys and none having any doubt as to the correctness of assignment as males. And many reports exist where, despite the absence of a normal penis^{19, 32, 58-62}, males were originally reassigned as girls, that switched back, and successfully lived as males. Several of these cases offer the same sorts of findings presented by us; with great similarity at the ages at which various milestones were passed, feelings developed and their reassignment challenged.^{59, 60} A most recent case illustrates this.

Reiner⁶³ reported on an adolescent Hmong immigrant who precipitously dropped out of school at 14 years of age. Upon subsequent interview she declared, although having been unequivocally raised as a girl from birth "I am not a girl, I am a boy". Indeed, physical examination revealed a 46 chromosome, XY male with mixed gonadal dysgenesis with a female appearing pelvis with clitoral hypertrophy. All her school-age friends had been boys. She enjoyed rough and tumble play, avoided dolls and girls activities and would dress in a gender-neutral or boys way. Her feelings of being different --being a boy-- developed from about the age of 8 and came to a head at 14 years. Treatment involved surgery

and endocrine therapy. This individual, after a period of some depression, progressively developed into a gynecophilic sexually active male.

These cases of successful gender change, as well as the present one, also challenge the belief that such a switch after the age of 2 years will be devastating. Indeed, in these cases it was salutary.

It must be acknowledged that cases of males accepting life as females after the destruction of their penises has been reported.⁶⁴ These reports, however, do not detail the individuals' sexual or personal lives.

Conclusions

Considering this case follow-up, and as far as an extensive literature review can attest, there is no known case where a 46 chromosome, XY male, unequivocally so at birth, has ever easily and fully accepted an imposed life as an androphilic female regardless of the physical and medical intervention. True, surgical reconstruction of traumatized male or ambiguous genitalia to that of a female, and attendant sex reassignment of males is mechanically easier than constructing a penis but there might be an unacceptable psychic price to pay. Concomitantly there is no support for the postulates that individuals are psychosexually neutral at birth or that healthy psychosexual development is dependent upon the appearance of the genitals. Certainly long term follow up on other cases is needed.

In the interim, however, new guidelines are offered. We believe that any 46-chromosome, XY individual born with a normal nervous system, in keeping with the psychosexual bias thus prenatally imposed, should be raised up as a male. Surgery to repair any genital problem, while difficult, should be conducted in keeping with this paradigm. This decision is not simple^{7, 13, 18, 63, 65-67} and analysis should continue.

As parents will still want their children to be and look normal as soon after birth as possible, physicians will have to provide the best advice and care, consistent with present knowledge. We suggest this means referring the parents and child to appropriate and periodic long term counseling rather than immediate surgery and sex reassignment just because that seems a simpler immediate solution to a complicated problem. With this management, a male's predisposition to act as a boy and his actual behaviors will be reinforced in daily interactions on all sexual levels and his fertility preserved. Social difficulties may reveal themselves as puberty is experienced, however, there is no evidence that with proper counseling and surgical repair when best indicated, an adjustment will not be managed as teen-agers manage other severe handicaps. Future reports will determine if we are correct.

REFERENCES

- ¹ Duckett JW, Baskin LS. "Genitoplasty for intersex anomalies". *European J Pediatrics*. 1993; 152 [Suppl 2]:580-584.
- ² Perlmutter AD, Reitelman C. "Surgical management of intersexuality". *Campbell's Urology*, 1992:1951-1966.
- ³ Behrman RE, Kliegman RM. *Nelson's Essentials of Pediatrics*. Philadelphia: W. B. Saunders Co., 1994.

- ⁴ Blethen SL, Weldon VV. "Disorders of External Genital Differentiation". In: Kelly VC, ed. *Practice of Pediatrics*. Vol. 7. Philadelphia: Harper & Row, 1985:1-23.
- ⁵ Catlin EA, Crawford JD. "Neonatal Endocrinology". In: Oski FA, ed. *Principles and Practice of Pediatrics*. Philadelphia: Lippincott, 1990:420-429.
- ⁶ Ratzan SK. "Endocrine & Metabolic Disorders". In: Dworkin PH, ed. *Pediatrics*. Baltimore: Williams & Wilkins, 1996:523-565.
- ⁷ Kessler SJ. "The medical construction of gender: case management of intersexed infants". *Signs: Journal of Women in Culture and Society* 1990; 16:3-26.
- ⁸ Money J, Hampson JG, Hampson JL. "An examination of some basic sexual concepts: the evidence of human hermaphroditism". *Bulletin of the Johns Hopkins Hospital* 1955; 97:301-319.
- ⁹ Money J. "Sex hormones and other variables in human eroticism". In: Young WC, ed. *Sex and Internal Secretions*. Baltimore: Williams and Wilkins, 1961:1383-1400.
- ¹⁰ Money J. "Cytogenetic and Psychosexual incongruities with a note on space form Blindness". *American Journal of Psychiatry* 1963; 119:820-827.
- ¹¹ Money J, Ehrhardt A. *Man and Woman, Boy and Girl*. Baltimore: John Hopkins University Press, 1972.
- ¹² Money J, Tucker P. *Sexual Signatures: On Being a Man or Woman*. Boston: Little, Brown, 1975.
- ¹³ Money J. *Sex errors of the body and related syndromes.: A guide to counseling children, adolescents, and their families*. Baltimore: Paul H. Brookes, 1994:132.
- ¹⁴ Burg FD, Merrill RE, Winter RJ, Schaible HD. *Treatment of infants, children and adolescents*. Philadelphia: W.B. Saunders, 1990.
- ¹⁵ Donahoe PK, Hendren WHI. "Evaluation of the Newborn with Ambiguous Genitalia". *Pediatric Clinics of North America* 1976; 23:361-370.
- ¹⁶ Diamond M. "Sexual Identity, Monozygotic Twins Reared in Discordant Sex Roles and a BBC Follow-Up". *Archives of Sexual Behavior* 1982; 11:181-185.
- ¹⁷ Savin-Williams RC. "Self-labeling and disclosure among gay, lesbian and bisexual youths". In: Green RJ, Laird J, eds. *Lesbians and gays in couples and families*. San Francisco: Jossey-Bass, 1996:153-182.
- ¹⁸ Diamond M. "Sexual Identity and Sexual Orientation in Children With Traumatized or Ambiguous Genitalia". *Journal of Sex Research* 1996e.
- ¹⁹ Cappon D, Ezrin C, Lynes P. "Psychosexual identification (psychogender) in the intersexed". *Canadian Psychiatric Association Journal* 1959; 4:90-106.
- ²⁰ Diamond M. "A critical evaluation of the ontogeny of human sexual behavior". *Quarterly Review of Biology* 1965; 40:147 - 175.

- ²¹ Roth M, Ball JRB. "Psychiatric Aspects of Intersexuality". In: Armstrong CN, Marshall AJ, eds. *Intersexuality: In vertebrates including man*. London: Academic Press, 1964:395-443.
- ²² Money J, Zuger B. "Critique and rebuttal". *Psychosomatic Medicine* 1970; 3: 463-467.
- ²³ Zuger B. "Gender role determination: a critical review of the evidence from hermaphroditism". *Psychosomatic Med.* 1970: 32:449-463.
- ²⁴ Stoller RJ. "The Intersexed Patient- Counsel and Management". In: Wahl CW, ed. *Sexual Problems: Diagnosis and Treatment in Medical Practice*. New York: The Free Press, 1967:149-162.
- ²⁵ Lev-Ran A. "Gender role differentiation in hermaphrodites". *Arch. Sex behav.* 1974;3:391.
- ²⁶ Money, J. "Hormones, hormonal anomalies, and psychological health-care". In: Kappy MS, Blizzard RM, Migeon CJ, eds. *Wilkin's Diagnosis and Treatment of Endocrine disorders in Childhood and Adolescence* 4th ed. Springfield Ill. Charles C. Thomas Publishers 1994;1141-1178.
- ²⁷ Stoller RJ. "The intersexed patient - counsel and management". In Wahl CW, ed. *Sexual Problems: Diagnosis and Treatment in Medical Practice*. New York: Free Press: 1967:149-162.
- ²⁸ Stoller RJ. *Sex and gender: On the Development on masculinity and femininity*. New York: Science House, 1968. 238-240.
- ²⁹ Diamond M. "Self-Testing: A Check on Sexual Levels". In: Bullough B, Bullough VL, eds. *Cross dressing and transgenderism*. Buffalo, New York: Prometheus, 1996b.
- ³⁰ Diamond M. "Self-Testing Among Transsexuals: A check on Sexual Identity". *Journal of Psychology & Human Sexuality* 1996; 8: in press.
- ³¹ Diamond M. "Some Genetic Considerations in the Development of Sexual Orientation". In: Haug M, Whalen RE, Aron C, Olsen KL, eds. *The Development of Sex Differences and Similarities in Behaviour*. Vol. 73. Dordrecht/Boston/London: Kluwer Academic Publishers, 1993:291-309.
- ³² Diamond M. "Biological Aspects of Sexual Orientation and Identity". In: Diamant L, McAnulty R, eds. *The Psychology of Sexual Orientation, Behavior and Identity: A Handbook*. Westport, Connecticut: Greenwood Press, 1995:45-80.
- ³³ Bailey JM, Pillard RC. "A genetic study of male sexual orientation". *Arch. Gen. Psychiat.* 1991; 48:1089-1096.
- ³⁴ Bailey JM, Pillard RC, Neale MC, Agyei Y. "Heritable Factors Influence Sexual Orientation in Women". *Arch. Gen. Psychiat.* 1993; 50:217-223.
- ³⁵ Bailey JM, Bell AP. "Familial aggregation of female sexual orientation". *Behavior Genetics* 1993; 23:312-322.
- ³⁶ Hamer DH, Hu S, Magnuson VL, Hu N, Pattatucci AML. "A Linkage Between DNA Markers on the X Chromosome and Male Sexual Orientation". *Science* 1993; 261:321-327.
- ³⁷ Pillard R, Poumadere J, Carretta R. "A Family Study of Sexual Orientation". *Archives of Sexual Behavior* 1982; 11:511-520.

- ³⁸ Pillard R, Weinrich J. "Evidence of Familial Nature of Male Homosexuality". *Archives of General Psychiatry* 1986; 43:808-812.
- ³⁹ Turner WJ. "Homosexuality, Type 1: An Xq28 phenomenon". *Archives of Sexual Behavior* 1995; 24:109-134.
- ⁴⁰ Whitam F, Diamond M, Martin J. "Homosexual Orientation In Twins: A report on 61 pairs and three triplet sets". *Archives of Sexual Behavior* 1993; 22:187-206.
- ⁴¹ Allen LS, Hines M, Shryne JE, Gorski RA. "Two sexually dimorphic cell groups in the human brain". *The Journal of Neuroscience* 1989; 9:497-506.
- ⁴² Allen LS, Gorski RA. Sexual orientation and the size of the anterior commissure in the human brain. *Proceedings of the National Academy of Sciences* 1992; 89: 7199-7202.
- ⁴³ Diamond M. "Perspectives in Reproduction and Sexual Behavior". In: Diamond M, ed. Bloomington: University of Indiana, 1968.
- ⁴⁴ Diamond M. "Sexual Identity and Sex Roles". In: Bullough V, ed. *The Frontiers of Sex Research*. Buffalo, N.Y.: Prometheus, 1979:33-56.
- ⁴⁵ Diamond M. "Bisexualit=E4t aus biologischer Sicht [Bisexuality: Biological Aspects]". In: Haeberle EJ, Gindorf R, eds. *Bisexualit=E4ten: Ideologie und Praxis des Sexualkontaktes mit beiden Geschlechtern [Bisexualities: Ideology and Practices of Sexual Contact with Both Sexes]*. Stuttgart: Gustav/Fischer Verlag, 1994:41-68.
- ⁴⁶ Gorski RA, Gordon JH, Shrayne JE, Southam AM. "Evidence for a morphological sex difference within the medial preoptic area for the rat brain". *Brain Research* 1978; 148:333-346.
- ⁴⁷ Gorski RA. "Hormone-induced sex differences in hypothalamic structure". *Bulletin Tokyo Metropolitan Institute of Neurosciences* 1988; 16:67-90.
- ⁴⁸ Hines M. "Gonadal hormones and human cognitive development". In: Balthazart J, ed. *Hormones, Brain and Behaviour in Vertebrates. 1. Sexual Differentiation, Neuroanatomical Aspects, Neurotransmitters and Neuropeptides*. Basel: Karger, 1990:51-63.
- ⁴⁹ Hines M. "Hormonal and neural correlates of sex-typed behavioral development in human beings". In: Haug M, Whalen RE, Aron C, Olsen KL, eds. *The development of sex differences and similarities in Behavior*. Dordrecht, The Netherlands: Kluwer Academic Publishers, 1993:131-149.
- ⁵⁰ LeVay S. "A difference in Hypothalamic Structure Between Heterosexual and Homosexual Men". *Science* 1991; 253:1034-1037.
- ⁵¹ LeVay S. *The Sexual Brain*. Boston: M.I.T. Press, 1993.
- ⁵² LeVay S, Hamer DH. "Evidence for a Biological Influence in Male Homosexuality". *Scientific American* 1994:44-49.
- ⁵³ Swaab DF, Fliers E. "A sexually dimorphic nucleus in the human brain". *Science* 1985; 228:1112-1115.

- ⁵⁴ Swaab DF, Hofman MA. "Sexual differentiation of the human hypothalamus: ontogeny of the sexually dimorphic nucleus of the preoptic area". *Developmental Brain Research* 1988; 44:314-318.
- ⁵⁵ Swaab DF, Hofman MA. "An enlarged suprachiasmatic nucleus in homosexual men". *Brain Research* 1990; 537:141-148.
- ⁵⁶ Swaab DF, Gooren LJG, Hofman MA. "Brain research, gender and sexual orientation". *Journal of Homosexuality* 1995; 28:283-301.
- ⁵⁷ Reilly JM, Woodhouse CRJ. "Small penis and the male sexual role". *The Journal of Urology* 1989; 142:569-572.
- ⁵⁸ Burns E, Segaloff A, Carrera GM. "Reassignment of Sex: report of 3 cases". *Journal of Urology* 1960; 84:126.
- ⁵⁹ Dicks GH, Childers AT. "The social transformation of a boy who had lived his first fourteen years as a girl: A case history". *American Journal of Orthopsychiatry* 1934; 4:508-517.
- ⁶⁰ Ghabrial F, Girgis SM. "Reorientation of sex: report of two cases". *International Journal of Fertility* 1962; 7:249-258.
- ⁶¹ Hoenig J. "The origins of gender identity". In: Steiner WB, ed. *Gender dysphoria: Development, research, management*. New York: Plenum Press, 1985:11-32.
- ⁶² Khupisco V. "The tragic boy who refuses to be turned into a girl". *Sunday Times of Johannesburg*. Johannesburg, 1995. 21 May.
- ⁶³ Reiner WG. "Case Study: Sex reassignment in a teenage girl". *Journal of American Academy of Child and Adolescent Psychiatry* 1996; 35:799-803.
- ⁶⁴ Gearhart JP. "Total ablation of the penis after circumcision with electrocautery: A method of management and long term followup". *Journal of Urology* 1989; 142:799-801.
- ⁶⁵ Fausto-Sterling A. "The Five Sexes: Why Male and Female Are Not Enough". *The Sciences* 1993; 1993:20-25.
- ⁶⁶ Meyer-Bahlburg HFL. Gender Identity Development in Intersex Patients. *Child and Adolescent Psychiatric Clinics of North America* 1993; 2:501-511.
- ⁶⁷ Zucker KJ, Bradley SJ. *Gender identity disorder and psychosexual problems in children and adolescents*. New York: The Guilford Press, 1995.